Requirement Analysis and Literature Survey

Project: Real-Time Sign Language Detection and Translation System

Functional Requirements

Functional requirements define the core capabilities the system must provide to detect, translate, and output sign language communication in real time.

- FR1 — Video Capture: The system shall capture continuous live video input from a connected webcam at a minimum of 30 frames per second.
- FR2 — Hand Detection: The system shall detect the presence of one or both hands within the camera frame and localize hand regions in real time.
- FR3 — Landmark Extraction: The system shall extract normalized hand-landmark coordinates (finger joints and palm key points) from each detected hand per frame.
- FR4 — Gesture Classification: The system shall classify static and dynamic hand gestures into corresponding signs using a trained CNN-LSTM model, from a supported vocabulary of at least 30–50 signs and the English alphabet.
- FR5 — Confidence Filtering: The system shall apply a configurable confidence threshold and discard or flag predictions below that threshold to reduce false recognitions.
- FR6 — Sentence Assembly: The system shall buffer sequential recognized signs and assemble them into readable words/sentences, handling repeated or held gestures appropriately.
- FR7 — Text Output: The system shall display the recognized text output on-screen in real time, updating as new signs are recognized.
- FR8 — Speech Synthesis: The system shall convert recognized text into audible speech using a text-to-speech engine.
- FR9 — Session Reset/Control: The system shall allow the user to clear the current sentence buffer, pause detection, and restart a session via simple on-screen or keyboard controls.
- FR10 — Model Update Support: The system shall support retraining or extending the gesture vocabulary without requiring a full redesign of the pipeline.

Non-Functional Requirements

Non-functional requirements specify the quality attributes the system must satisfy to be practical, dependable, and usable in real-world settings.

Quality Attribute	Requirement
Performance	The system shall process video frames and return gesture predictions with an end-to-end latency of under 300 ms per sign, sustaining near real-time interaction on standard laptopclass CPU hardware.

Reliability	The system shall maintain a gesture classification accuracy of at least 90% on a held-out validation set and degrade gracefully (e.g., flag low-confidence input) rather than fail silently under poor lighting or partial hand occlusion.
Usability	The interface shall present recognized text and controls in a simple, uncluttered layout
	usable by individuals without technical training, requiring no more than basic on-screen guidance to operate.
Scalability	The system architecture shall support extension of the gesture vocabulary and future adaptation to additional sign languages (e.g., regional sign languages) without rearchitecting the core pipeline.
Security & Privacy	Video frames shall be processed locally in real time and shall not be stored or transmitted beyond what is required for immediate inference, protecting user privacy; any optional cloud TTS calls shall use secure (HTTPS) connections.
Maintainability	The codebase shall be modular (separating capture, detection, classification, and output stages) with clear interfaces, enabling individual components to be updated, tested, or replaced independently.
Portability	The system shall run on common desktop operating systems (Windows, Linux, macOS) using only a standard webcam, without dependency on specialized proprietary hardware.
Availability	The application shall be able to run as a standalone offline tool for the core detection and text-output functions, with cloud dependency limited to optional enhanced speech synthesis.

Literature Survey

Research Papers and Existing Solutions

The table below reviews representative research papers and existing systems relevant to automated sign language recognition and translation, spanning classical statistical methods, deep-learning approaches, and a commercial solution.

S.No.	Research Paper / Source	Technology Used	Approach Summary	Limitation
1	Starner & Pentland (1995)	Hidden Markov Models, coloredglove tracking	Recognized ASL sentences from video using HMMs on tracked hand features.	Required colored gloves; limited vocabulary; not robust to unconstrained bare-hand input.
2	Pigou et al. (2015)	CNNs on RGB-D (Kinect) video	Applied deep CNNs to depth video for Italian gesture recognition.	Dependent on depth-sensing hardware, restricting deployment to ordinary cameras.
3	Koller et al. (2016)	CNN + HMM hybrid	Weakly-supervised continuous sign recognition on the RWTH-PHOENIXWeather corpus.	High computational cost; relies on large domainspecific annotated corpora.

4	Camgoz et al. (2018)	Encoder-decoder neural translation	Maps sign video directly to spoken-language text end-toend.	Needs substantial training data and GPU resources; not lightweight for real-time use.
5	Rastgoo et al. (2021)	Survey: CNN, RNN/LSTM, hybrid models	Comparative survey of deeplearning sign recognition methods across datasets.	Notes most systems are evaluated offline, not optimized for real-time/lowresource inference.
6	Li et al. — WLASL Dataset (2020)	I3D video-based recognition	Introduced a large word-level ASL video dataset and	Benchmark models are computationally heavy for
			benchmark models.	real-time, on-device deployment.
7	SignAll (commercial system)	Multi-camera array + glove-based tracking	Commercial ASL-to-text translation system using specialized capture hardware.	High hardware cost and installation complexity; not suited for casual, low-cost personal use.
8	Google MediaPipe Hands (2020)	Lightweight ondevice handlandmark model	Provides real-time, bare-hand landmark tracking usable on consumer devices without extra hardware.	A perception building block only — provides landmarks but no gesture classification or translation on its own.

Technologies Used in Existing Work

- Statistical sequence models: Hidden Markov Models (HMMs) for early gesture and sign sequence recognition.
- Depth-sensing hardware: Microsoft Kinect and similar RGB-D cameras for 3D hand/body tracking.
- Deep learning architectures: Convolutional Neural Networks (CNNs) for spatial feature extraction; Recurrent Neural Networks / LSTMs for temporal sequence modeling; encoder-decoder and I3D-based architectures for end-to-end video-to-text translation.
- Specialized capture hardware: instrumented gloves and multi-camera arrays (e.g., commercial systems such as SignAll) for high-precision hand tracking.
- Lightweight on-device perception frameworks: MediaPipe and similar libraries providing real-time handlandmark detection from ordinary RGB video.
- Large annotated datasets: RWTH-PHOENIX-Weather, WLASL, and similar corpora used to train and benchmark recognition models.

Limitations of Existing Systems

- Hardware dependency: A large share of high-accuracy systems require specialized equipment (depth cameras, sensor gloves, multi-camera rigs), making them expensive and impractical for everyday personal use.
- Computational cost: State-of-the-art deep architectures (encoder-decoder translation models, I3D-based classifiers) demand significant GPU resources, limiting real-time deployment on ordinary consumer devices.

- Vocabulary and generalization limits: Many systems are trained and validated on narrow, domain-specific vocabularies (e.g., weather-broadcast signing) and do not generalize well to everyday, general-purpose signing.
- Offline evaluation focus: Much of the published research reports accuracy under offline, batch-processing conditions rather than true real-time, interactive constraints.
- Cost and accessibility: Commercial solutions with strong accuracy are often costly to install and maintain, putting them out of reach for individual users, schools, or small service providers.

Research Gap Identified

Across the reviewed literature, a consistent gap emerges between accuracy and accessibility: systems that achieve the highest recognition accuracy typically depend on specialized sensors or heavy computational infrastructure, while lightweight, camera-only systems tend to support only small, isolated-gesture vocabularies and are rarely evaluated for real-time responsiveness. There is a clear gap for a solution that combines bare-hand, RGB-camera-only input with an efficient model architecture capable of real-time performance on consumer-grade hardware, while still supporting a practically useful vocabulary and continuous sentence-level output — without requiring costly specialized equipment.

How the Proposed Project Addresses This Gap

- Uses only a standard RGB webcam and a lightweight, pre-trained hand-landmark extractor (e.g., MediaPipe), eliminating the need for gloves, depth sensors, or multi-camera rigs.
- Employs a compact CNN-LSTM classifier operating on landmark features rather than raw video frames, reducing computational load enough to run in real time on standard laptop-class CPUs.
- Targets an explicit real-time latency budget (under 300 ms per sign) and reports it as a measured outcome, directly addressing the offline-evaluation gap identified in prior surveys.
- Includes sentence-level buffering and assembly logic so that recognized signs are combined into coherent, continuous output rather than isolated single-gesture classifications.
- Is designed for low-cost, everyday deployment (a webcam and a laptop), making it accessible to individuals, schools, and small service providers who cannot afford specialized commercial systems.